

HW04 - AI Automation Testing Report

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<https://github.com/linhnph05/hw04---Automation-testing> Date: 2026-08-09

1. Scope and feature selection

I automated these three web features:

- Pool A - FR-03: Forgot password and reset password.
- Pool B - FR-09: Discount coupons.
- Pool C - FR-12: Admin access control.

2. Method

I used a staged AI-first workflow. I first asked an Codex agent to inspect the requirements and source. Next, I asked it to design exactly 12 cases per feature. Only after reviewing those cases did I request a Playwright draft for one feature at a time. I converted the drafts from CommonJS to the repository's ES module format, corrected weak assumptions, ran a Chromium baseline, asked the agent to critique its own drafts, and then made a final human-review pass.

The course notes recommend data-driven tests, role-based locators, web-first assertions, isolated state, API setup, and multi-browser execution. The final suite follows these points:

- Data is loaded from three external JSON files under `test-data/`.
- UI tests use Playwright role, placeholder, and form locators.
- API setup is used for fixture users and security checks.
- No fixed sleeps are used in the tests.
- Unique fixture IDs and cleanup prevent tests from depending on execution order.
- Each feature runs separately in Chromium, Firefox, and WebKit.

3. Automated cases and assertion patterns

Each JSON file contains 12 rows, giving 36 test cases. Across three browser projects, this produces 108 executions.

All 36 cases are also documented with their preconditions, input, actions, and expected result in `TEST_CASES.md` (`TEST_CASES.md`). This is the human-readable text version of the test design; the JSON files remain the data sources used by Playwright.

The suite uses more than three assertion patterns, including:

1. Visibility and count: `toBeVisible()` and `toHaveCount()`.
2. Exact values and status codes: `toBe()`.
3. Text and regular expressions: `toMatch()` and `toContainText()`.
4. Navigation: `toHaveURL()`.
5. Object structure: `toMatchObject()`.
6. Negative state checks: `not.toEqual()` and verification that unauthorized fixtures were not stored.

4. Execution results

Feature	Chromium	Firefox	WebKit	Total passed	Total failed
FR-03	7 passed / 5 failed	7 / 5	7 / 5	21	15
FR-09	7 passed / 5 failed	7 / 5	7 / 5	21	15
FR-12	5 passed / 7 failed	5 / 7	5 / 7	15	21
Total	19 / 17	19 / 17	19 / 17	57	51

The failures are consistent across all three engines. This consistency and the requirement-based assertions indicate SUT defects rather than browser-specific flakiness. Each of the nine HTML reports was opened with Chromium and checked for a visible Run by: 23127081 title and ISO timestamp.

5. Human review of AI-generated drafts

AI draft problem	Human correction	Why the AI missed it
Used CommonJS require in an ESM project	Converted all drafts to import and file-based JSON parsing	The model followed common Playwright examples without checking project configuration
FR-03 TC-01 mixed OTP length with step indicators	Kept OTP format as the single primary assertion; TC-02 owns navigation controls	The model tried to reduce repeated setup and combined requirements
Relied on alert wording for reset results	Accepted dialogs without using their exact text and verified URL/login state	The model copied the current UI instead of testing the stable business result
Guest coupon lookup matched the permanent login link	Replaced the broad text poll with a direct login-URL expectation	The locator was not scoped to the coupon result area
Pretended a deleted coupon represented an inactive coupon	Replaced it with a real empty-code case and documented the inactive-fixture gap	No public API can create <code>is_active = 0</code>
Sent <code>user_id</code> on normal coupon calls	Normal calls now rely on JWT identity; only the spoofing case sends another ID	The model followed the buggy client contract too closely
Mixed per-user isolation with broken percent arithmetic	TC-11 now asserts successful isolation only; TC-01 owns exact percent math	One case had two independent failure reasons
FR-12 mutation tests checked only HTTP status	Added state checks proving unauthorized products, categories, or coupons were created	The first draft focused on the access response but not its side effect
Fixture names used only <code>Date.now()</code>	Added counters to prevent same-millisecond collisions	The model underestimated repeated setup speed with one worker

6. Defects found

Eight root-cause issues were created on GitHub. They cover all repeated assertion failures without duplicating one issue for every browser:

- FR-03: four-digit OTP; missing step/navigation/confirmation controls; valid password rejected.
- FR-09: incorrect percent math; wrong equality boundary; missing authentication and trusted body user ID.
- FR-12: missing admin-role checks; unauthenticated product write APIs.

Full mappings, evidence, and issue links are in `BUG_REPORT.md` (`BUG_REPORT.md`).

7. Automation gap

The exact inactive-coupon branch (`is_active = 0`) could not be isolated through the available UI or API because the SUT exposes no supported way to create or deactivate that fixture. I did not modify SQLite directly because that would couple the web automation to SUT internals. The suite still tests nonexistent, empty, expired, below-minimum, equality, usage-limit, guest, and identity-spoofing conditions. A future test-fixture endpoint or seed option should add the inactive case.

8. Report evidence

- HTML reports: `reports/html/<feature>-<browser>/index.html`.
- Visual summaries: `evidence/*-chromium-report.png`.
- Every failure retains a Playwright trace and screenshot during execution; the committed HTML reports contain the final run details.

9. Git history limitation

There are more than eight meaningful commits that modify `.spec.js` files. However, those HW04 test commits were made on 2026-08-09 rather than across four days. I did not falsify Git author dates. The repository has older history on another date, but it does not satisfy the rule that the qualifying test-script commits themselves span four days.

10. Conclusion

The final suite is data-driven, uses multiple assertion styles, runs on three browser engines, and produces nine attributable reports. The stable cross-browser failures found important correctness and authorization defects. AI accelerated source analysis and draft generation, but human review was necessary to correct module format, selectors, fixture assumptions, assertion scope, and cleanup.